

Master Team Project Summer 2025

WWW site for buy/sell/exchange
exclusively for Fulda students, staff and
faculty

UniMarkt

Milestone 1

Team 5

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1. Executive Summary

UniMarkt is envisioned as an intuitive, secure, and campus-focused online marketplace that bridges the gap between Fulda University's students, faculty, and staff in managing surplus goods and services. Recognizing the limitations of broad commercial platforms where irrelevant listings, uncertain seller credibility, and opaque transaction processes can overwhelm users, our platform narrows the focus to the HS-Fulda community. By requiring registration with an official university email, UniMarkt creates a trusted environment where every user is verified and accountable, fostering confidence in both listing and acquisition activities.

At its core, UniMarkt streamlines the lifecycle of campus items, from textbooks and electronics to furniture and specialty services such as language tutoring or gym membership transfers. Sellers can describe their offerings in rich, structured forms, set conditions and prices, and indicate whether items are for sale, exchange, lending, or free pickup. Buyers benefit from a responsive interface that adapts to list and grid views, lets them filter by category, price range, proximity, and item condition, and highlights trusted sellers based on community ratings. Behind the scenes, a dedicated administrator moderates new listings for compliance with university policies and ensures that content remains appropriate and safe for all participants.

What truly distinguishes UniMarkt is its commitment to sustainability, collaboration, and local relevance. By encouraging the reuse and sharing of resources within a bounded community, the platform reduces waste and fosters interpersonal connections among peers. In-platform messaging keeps negotiations and arrangements confidential until both parties are ready to exchange contact details, while an integrated feedback system and trust scores cultivate a culture of reliability. Additional conveniences such as map-based location display, timestamped "new" badges, and a consolidated cart for multiple reservations add to the efficiency and appeal of the experience.

Our initiative is driven by a dedicated team of five Global Software Development students at HS-Fulda, each bringing specialized expertise to the project. Mohammed Zikrullah serves as Team Lead and Front-End Lead, overseeing the user interface and working closely with Murad Isayev on all front-end development. Aman Aneet Pandey holds the role of Back-End Lead, responsible for architecting and implementing our FastAPI services and MySQL database. Gaurav Bhaskar Bhamre collaborates with Aman on the FastAPI development, ensuring seamless integration between services. Faizan Anwar supports the entire team as GitHub Master, managing version control, CI/CD pipelines, and code quality. Together, we are committed to delivering a robust, scalable platform that transforms underutilized campus resources into cost-effective, community-driven opportunities.

2.1 Personas

Persona 1: Student Seller (Sarah)

Sarah is a 22-year-old Computer Science student at Fulda University who needs to manage her limited budget effectively. She frequently has textbooks, electronics, and dormitory items she no longer needs after completing courses or upgrading her equipment. With limited storage space in her small student apartment, she wants to declutter by selling items she no longer uses. Sarah is tech-savvy but has limited time between classes, study sessions, and part-time work. She finds existing marketplace platforms overwhelming with too many irrelevant listings from outside the university community. She values safety and convenience when selling items and prefers dealing with fellow students who understand campus logistics.

Persona 2: Faculty Buyer (Professor Weber)

Professor Weber is a 45-year-old faculty member in the Engineering department who has been teaching at Fulda for over 10 years. He frequently needs specialized academic materials, office equipment, and occasionally teaching resources that the university procurement process takes too long to provide. He has a moderate technical ability but prefers straightforward interfaces that don't require a learning curve. His pain points include limited departmental budgets for equipment and difficulty finding specific items quickly through traditional retail channels. He values quality items at reasonable prices and appreciates the convenience of on-campus exchanges to fit his busy schedule.

Persona 3: International Student User (Miguel)

Miguel is a 24-year-old exchange student from Spain studying International Business at Fulda for one semester. With limited finances and temporary housing, he needs to acquire essential items like furniture, kitchen supplies, and winter clothing without investing heavily in things he can't take home. He also wants to exchange language tutoring services for help with German language practice. His technical skills are good, but he sometimes struggles with complex German instructions. Miguel's pain points include the high cost of new items for his temporary stay and difficulties navigating the local second hand markets due to language barriers. He values

community connections and prefers dealing with fellow university members who might be more understanding of his situation.

Persona 4: Staff Administrator (Martina)

Martina is a 38-year-old administrative staff member in the Student Affairs department. She has been tasked with moderating the university's new buy/sell/exchange platform. With intermediate technical skills but strong organizational abilities, she needs to efficiently review listings, handle user concerns, and maintain platform integrity. Her pain points include balancing moderation responsibilities with her regular administrative duties and identifying potentially problematic listings quickly. She values clear guidelines for moderation decisions and straightforward tools that allow her to efficiently manage the platform with minimal time investment.

2.2 Use Cases

Use Case 1: Listing an Item for Sale

A student user like Sarah wants to sell her used calculus textbook after completing her mathematics course. After logging into the platform using her university credentials, she creates a new listing by uploading clear photos of the book, providing details about its condition (highlighting minimal highlighting and intact binding), setting a reasonable price based on similar listings, and selecting relevant categories and tags. She specifies her preferred handover locations on campus and available time slots between her classes. After submitting the listing, she waits for approval from a moderator before her item becomes visible to potential buyers within the university community.

Use Case 2: Searching and Filtering Available Items

Professor Weber needs to find an affordable digital projector for his department's upcoming conference. He logs into the platform and uses the search function with specific criteria: electronics category, projector subcategory, under €200, and preferably available for pickup in the next three days. The system displays relevant results from other faculty, staff, and students selling suitable projectors. He browses through the filtered results, comparing prices, conditions, and specifications of available projectors. After finding a promising option, he can view detailed

information about the item and the seller's university affiliation before deciding to contact them through the platform's messaging system.

Use Case 3: Coordinating Exchange Through In-Platform Messaging

After finding a desired item, a staff member initiates contact with the seller through the platform's secure messaging system. They discuss item details, negotiate on price if appropriate, and arrange a mutually convenient time and campus location to meet for the exchange. The messaging system allows them to share contact information only when both parties feel comfortable proceeding with the transaction. After meeting in person to inspect the item and complete the payment (handled outside the platform), both parties return to mark the transaction as complete. They can provide feedback about their experience, which contributes to their respective reputation scores within the university marketplace community. The system automatically archives their conversation and updates the listing status to "Sold," maintaining a record of the transaction for reference.

Use Case 4: Exchanging Services Among University Members

Miguel wants to offer Spanish language tutoring in exchange for help with his German assignments. He creates a service exchange listing specifying his offering (2 hours of Spanish conversation practice per week) and what he seeks in return (1 hour of German homework review). He categorizes this under "Academic Services" and adds relevant tags. After approval, German-speaking students interested in learning Spanish can browse his listing and contact him through the platform's messaging system to discuss logistics. They negotiate meeting times and locations on campus that work for both parties, without needing to share personal contact information until they're comfortable proceeding with the exchange.

Use Case 5: Moderating Platform Content

Martina logs into her administrator account to review newly submitted listings awaiting approval. She sees a queue of items and services that users have posted but are not yet visible to the community. She reviews each listing according to university guidelines, checking for prohibited items, inappropriate content, reasonable pricing, and complete information. For acceptable listings, she approves them with a single click, making them immediately visible to

the university community. For problematic listings (e.g., a textbook with excessive pricing or incomplete information), she selects from standardized rejection reasons and adds personalized guidance for the seller on how to correct their listing for future approval.

3. List of main data items and entities

3.1 User Entity

Description: Represents a registered individual using the platform.

Attributes	Details
UserID	Unique identifier for each user (internal reference).
Name	Full name of the user.
Email	Fulda University email address (must be verified).
Password	Hashed user password.
UserRole	One of Student, Staff, or Faculty.
ProfilePicture	Optional image to represent the user.
PhoneNumber	Optional contact number for offline communication.
RegistrationDate	Date and time when the user joined.

3.2 Listing Entity

Description: A product or service a user wants to sell, buy, or exchange.

Attributes	Details
ListingID	Unique identifier for each listing.
Title	Short descriptive title.
Description	Detailed information about the item/service.
Category	Category (e.g., Electronics, Books, Furniture, Services, Housing).
ListingType	One of Sell, Buy, or Exchange.
Price	Price in euros (or "Negotiable" tag).
Images	One or more optional images.
Location	Optional location/area within or near Fulda.
Status	One of Active, Sold, Archived.
PostedBy	Link to UserID of the creator.
PostedDate	Timestamp of creation.

Behavior

- **Expiration:**
 - Listings automatically expire after **30 days** unless renewed.
 - Example: A listing posted on **2025-01-01** expires on **2025-01-30** if not renewed.
- **User Permissions:**
 - Users can **update** their listings (e.g., edit price, description, or images).
 - Users can **deactivate** listings (marked as Sold or Archived).

3.3 Message/Chat Entity

Description: Represents private messages between two users regarding a listing.

Attributes	Details
MessageID	Unique ID for the message.
SenderID	UserID of the sender.
ReceiverID	UserID of the recipient.
ListingID	(Optional) If related to a specific listing.
MessageContent	Text of the message.
Timestamp	Date and time of message.
ReadStatus	Read or Unread.

3.4 Report Entity

Description: Allows users to report listings or messages violating rules.

Attributes	Details
ReportID	Unique ID for the report.
ReportedBy	UserID who initiated the report.
TargetType	One of Listing or User.
TargetID	ID of the item/user being reported.
Reason	Predefined reason (e.g., Spam, Scam, Inappropriate Content).
AdditionalNotes	Optional detailed explanation.
ReportDate	Timestamp.

Attributes	Details
Status	One of Open, Under Review, Resolved.

3.5 Category Entity

Description: Used to classify listings into groups.

Attributes	Details
CategoryID	Unique identifier.
Name	Name of category (e.g., Electronics, Books).

3.6 Admin Entity

Description: Special system user responsible for moderating the platform.

Attributes	Details
AdminID	Unique identifier.
Name	Full name.
Email	Admin email.
Password	Encrypted password.

Privileges

- **User & Listing Management:**
 - View and manage any user account or listing.
 - Access full user activity logs (e.g., registration date,email id).
- **Report Resolution:**
 - Investigate and resolve user-generated reports (e.g., scams, inappropriate content).
 - Escalate severe cases to higher authorities if needed.
- **Content Moderation:**

- Remove listings or comments violating platform guidelines.
- Block users temporarily or permanently for misconduct.
- **System Oversight:**
 - Adjust platform-wide settings (e.g., listing expiration time, categories).

4. Initial functional requirements

1. **User registration:** A user should be able to register by providing the required information.
2. **Exclusive Access :** Only users with a valid [@hs-fulda.de](mailto:hs-fulda.de) email should be able to register.
3. **User login / logout :** A registered user should be able to log in and log out from the website.
4. **Listing Approval :** An admin should be able to review and approve new item listings.
5. **Content moderation :** An admin should be able to remove inappropriate listings.
6. **Platform Monitoring :** An admin should be able to suspend or delete users accounts after receiving report cases and be able to moderate messages and activities.
7. **Search Products:** A user should be able to search products using keywords.
8. **Item browsing :** A user should be able to browse all available items after logging in.
9. **View options :** A user should be able to view items in both list and grid formats.
10. **Category Browsing:** A user should be able to filter items by selecting predefined categories.
11. **Sorting Options:** A user should be able to sort items by price, distance, popularity, or seller rating.
12. **Product filtering:** A user should be able to filter items based on price range / distance / popularity / rating / condition of product, etc.
13. **New Item Tags:** Newly added items should be marked with **New** tag.
14. **Create advertisement :** A seller should be able to create an advertisement by adding item photos, details, and all required details.
15. **Location sharing :** A seller should be able to optionally share their location using maps.
16. **Product listing :** A user should be able to view items available for purchase, borrow, or exchange.

- 17. Gym subscription transfer:** A user can sell gym subscription on the platform.
- 18. In-platform messaging :** Buyers and sellers can communicate via a built-in messaging system.
- 19. Responsive user interface :** The platform must be fully responsive and optimized for Desktop, Tablet, and mobile devices.
- 20. User cart :** User should be able to add items to the cart.
- 21. Maps :** Buyer should be able to see the location of the seller.
- 22. Related products :** Users should be able to see all similar products below the advertisement.
- 23. Advertisement sharing :** Users should be able to share the link for the advertisement.
- 24. Exchange / Borrow review form :** After the borrower returns the item he borrowed the lender will be asked about the experience.
- 25. Trusted tag :** Based on the review a user should be tagged trusted.
- 26. Hashing based user credential storage :** Users credentials will be stored in a hash based database for maintaining user privacy.
- 27. Top spot seller :** Top 1 seller will be highlighted on the top of the listing.
- 28. Buyer / Borrow Advertisement :** A user can put an advertisement based on his needs. He should be able to create a borrow or a buy advertisement.

5. List of non-functional requirements

- 1. Application shall be developed, tested and deployed using tools and servers approved by Class CTO and as agreed in Milestone 0. Application delivery shall be from chosen cloud server.
- 2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers.
- 3. All or selected application functions must render well on mobile devices.
- 4. Data shall be stored in the database on the team's deployment cloud server.
- 5. No more than 50 concurrent users shall be accessing the application at any time.
- 6. Privacy of users shall be protected and all privacy policies will be appropriately communicated to the users.

7. The language used shall be English (no localization needed).
8. Application shall be very easy to use and intuitive.
9. Application should follow established architecture patterns.
10. Application code and its repository shall be easy to inspect and maintain.
11. Google analytics shall be used (optional for Fulda teams).
12. No email clients shall be allowed.
13. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
14. Site security: basic best practices shall be applied (as covered in the class) for main data items.
15. Application shall be media rich (images, video etc.). Media formats shall be standard as used in the market today.
16. Modern SE processes and practices shall be used as specified in the class, including collaborative and continuous SW development.
17. For code development and management, as well as documentation like formal milestones required in the class, each team shall use their own GitHub to be set-up by class instructors and started by each team during Milestone 0.
18. The application UI (WWW and mobile) shall prominently display the following exact text on all pages *"Fulda University of Applied Sciences Software Engineering Project, Summer 2025 For Demonstration Only"* at the top of the WWW page. (Important so as to not confuse this with a real application).

6. Competitive Analysis

Products Features	eBay	Vinted	Kleinanzeigen	Our App
<i>In-platform Messaging</i>	✓	✓	✓	✓
<i>Seller rating system</i>	✓	✓	✓	✓
<i>Give-away items</i>	✗	✓	✓	✓
<i>Top seller highlight</i>	✓	✓	✗	✓
<i>Borrow / lend option</i>	✗	✗	✗	✓
<i>Gym subscription transfer</i>	✗	✗	✗	✓
<i>Exchange option</i>	✗	✗	✗	✓

Our App clearly stands out by offering all the listed features, including unique options like borrow/lend, gym subscription transfer, and exchange - none of which are available on eBay, Vinted, or Kleinanzeigen. While competitors cover core functions like messaging, seller ratings, and giveaways, Our App combines these with additional flexible options that meet more user needs, making it a more versatile and attractive platform.

7. High-level system architecture and technologies used

- **Server and Hosting**
 - Server host : Google Compute Engine (1 vCPU, 2 gb RAM)
 - Operating System : Ubuntu 22.04 LTS
- **Database**
 - Database : MySQL
- **Web Server**
 - Web server : Nginx 1.24.0
- **Server side**
 - Programming language : Python (v3.12)
 - Backend Framework : FastAPI (v0.111.0)

- **Frontend**
 - Framework : [React.js](https://reactjs.org/) (v18.2.0)
 - Styling language : CSS3 | SASS/SCSS (v1.87.0)
 - Styling framework : Tailwind (v4)
- **Tools**
 - Personal meeting : Slack
 - Project management : Trello
- **Package management** : NPM (v.11.3.0)
- **IDE** : Visual Studio Code and IntelliJ

8. Team and roles

Name	E-Mail	Role
Mohammed Zikrullah	gaurav-bhaskar.bhamare@informatik.hs-fulda.de	Team Lead / Front end lead
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Aman Aneet Pandey	aman-aneet.pandey@informatik.hs-fulda.de	Back end lead
Gaurav Bhaskar Bhamre	gaurav-bhaskar.bhamare@informatik.hs-fulda.de	Back end

9. Checklist

Team found a time slot to meet (online) outside of the class.	DONE
GitHub master chosen.	DONE
Team decided and agreed together on using the listed SW tools and deployment server.	DONE
Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing.	DONE
Team lead ensured that all team members read the final M1 and agree/understand it before submission	DONE
GitHub organized as discussed in class (e.g. master branch, development branch, folder for milestone documents etc.)	DONE